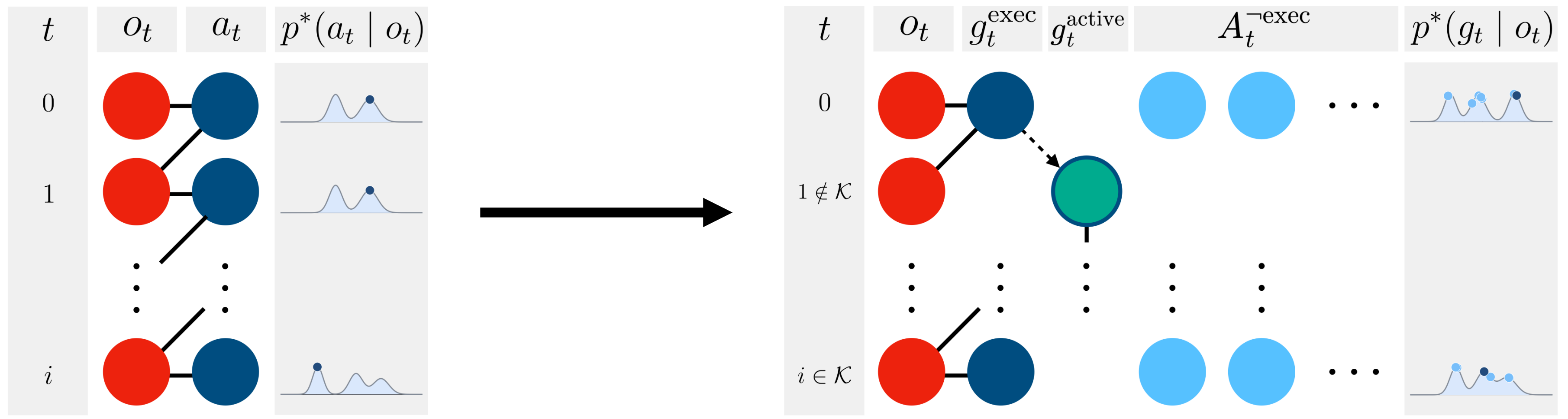
CrowdMAPS: Crowdsourcing Multiple Actions per State for Real-World Robot Imitation Learning

Yilong Song, Harshvardhan Chandirasekar, Karthik Desingh, and Chen Zhu-Tian
Department of Computer Science & Engineering
University of Minnesota, Twin Cities

Conference on Robot Learning (CoRL) 2026

Missing Supervision?

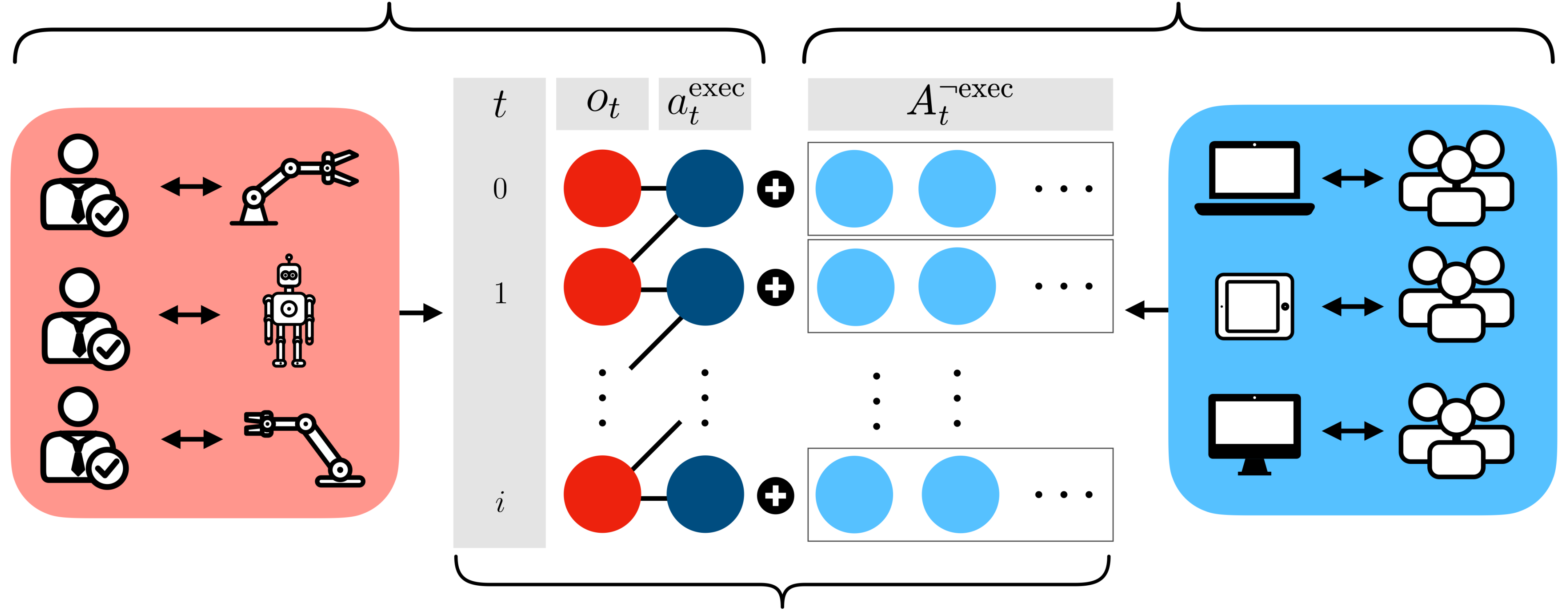
Every new robot trajectory requires physical interaction.
Yet each visited state often admits **many valid actions**—not just the one executed. Standard demonstrations record only one of them.


t
o_t
a_t
$p^*(a_t \mid o_t)$
0
1
i
t
o_t
g_t^{exec}
g_t^{active}
$A_t^{\neg\text{exec}}$
$p^*(g_t \mid o_t)$
0
$1 \notin \mathcal{K}$
$i \in \mathcal{K}$

Action Densification: A New Scaling Axis

Scale robot data in width, not just length.
Length: More trajectories, more physically visited states.
Width: More valid actions for states already visited.
MAPS densifies action supervision without adding physical state transitions.

(a) Real-Robot Data Collection (b) CrowdMAPS


t
o_t
a_t^{exec}
$A_t^{\neg\text{exec}}$
0
1
i

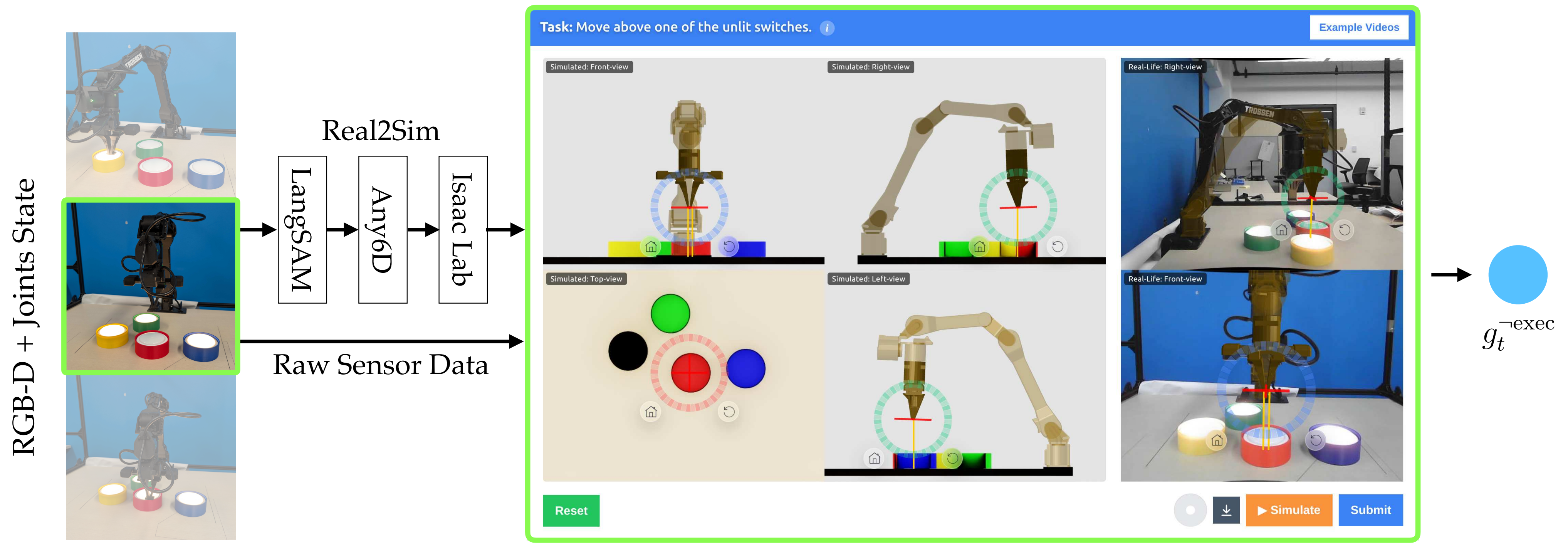
(c) Real-Robot MAPS Data

Decoupling Robots Hardware from Data Collection

Physical data collection acquires states. Remote contributors annotate alternative actions.

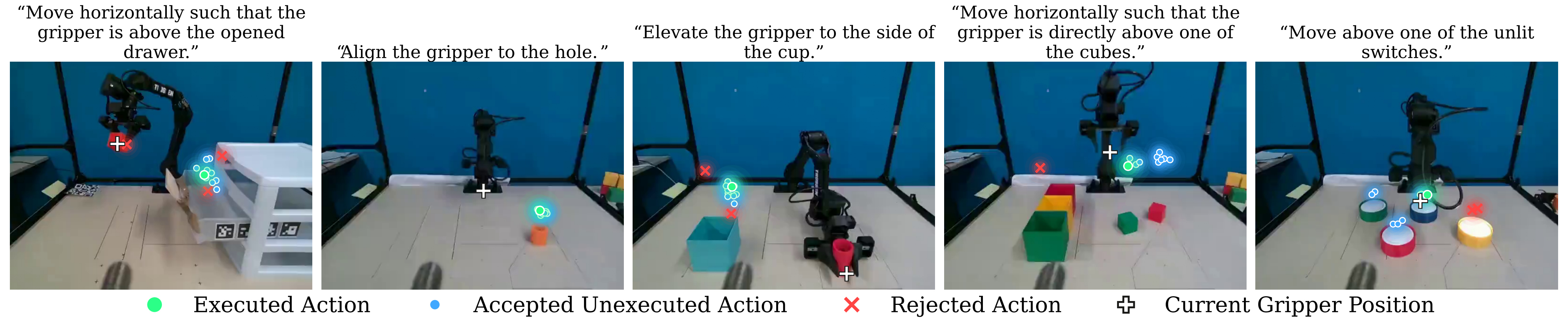
Physical Collection → **Reconstruct** → **Annotate** → **Validate**

Asynchronous annotation scales independently of physical robot execution.


RGB-D + Joints State
Real2Sim
LangSAM
Any6D
Isaac Lab
Raw Sensor Data
Task: Move above one of the unlit switches.
Example Videos
Simulated: Front-view
Simulated: Right-view
Real-Life: Right-view
Simulated: Top-view
Simulated: Left-view
Real-Life: Front-view
Reset
▶ Simulate
Submit
$g_t^{\neg\text{exec}}$

(a) Recorded Keyframes (b) Annotation UI

47 participants. 78.7% with no prior robot-arm experience.
7–12 s median submission time (four tasks with logs).
Submissions validated for task consistency and physical coherence.


"Move horizontally such that the gripper is above the opened drawer."
"Align the gripper to the hole."
"Elevate the gripper to the side of the cup."
"Move horizontally such that the gripper is directly above one of the cubes."
"Move above one of the unlit switches."
● Executed Action
● Accepted Unexecuted Action
✕ Rejected Action
✚ Current Gripper Position

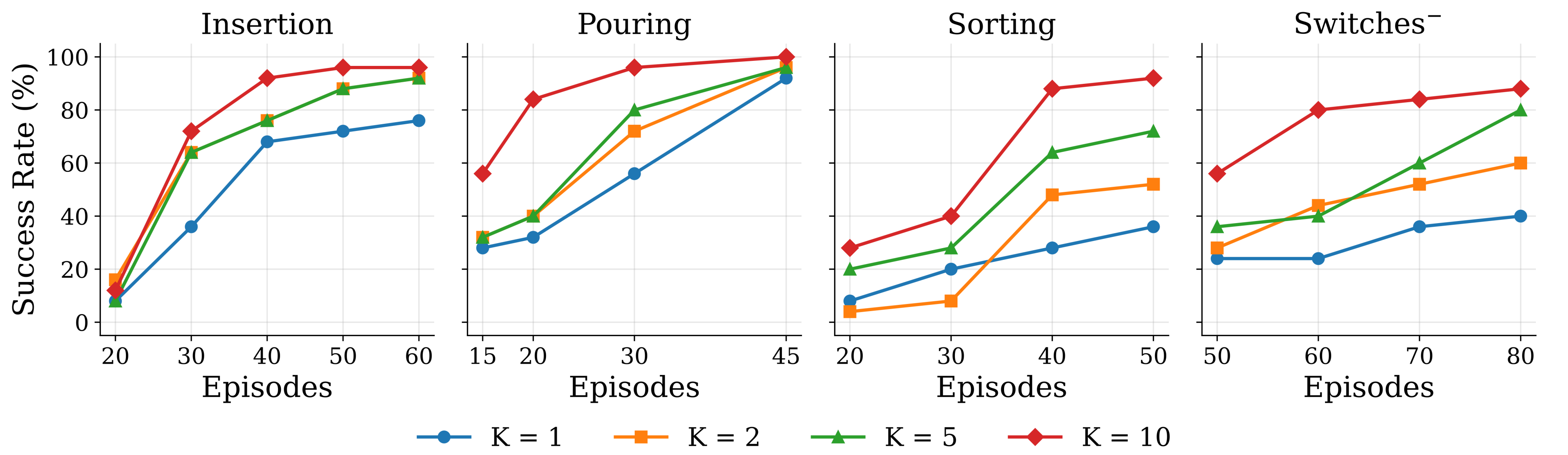
Executed and Crowd-Submitted Actions from Selected Keyframes

MAPS Data Improves Downstream Policies

Keypose policies

+35.6 pp

38.7% → 74.2% success · K = 1 → 10

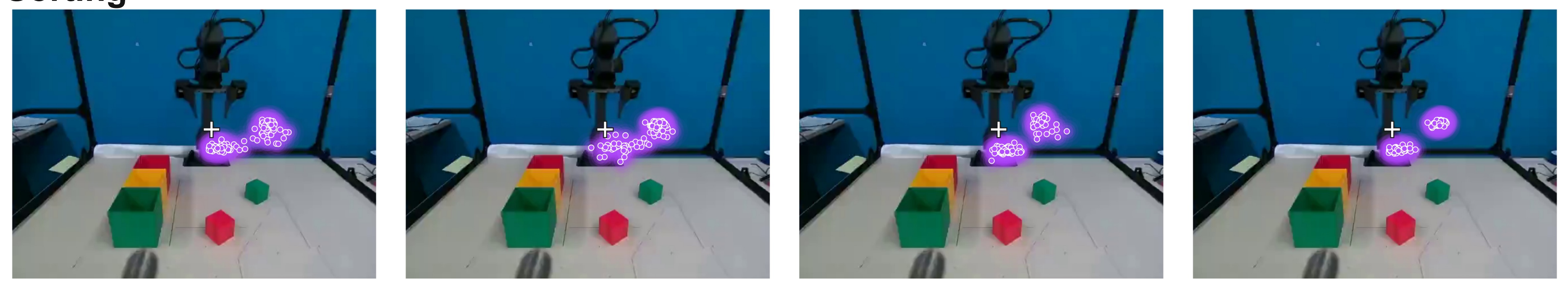


Higher success with fewer robot demonstrations.

Wider MAPS sharpens policy predictions.
Held-out states · K = 1, 2, 5, 10 (left to right)

Insertion

Sorting



Sharper goal estimates; better-separated action modes.

Trajectory policies

+33.0 pts

Average task performance · K = 1 → 10

At K = 10, MAPS outperforms extra expert trajectories on all five tasks.

Action densification makes every physical trajectory more valuable: more valid actions per visited state, more effective supervision, and better real-robot policies without proportionally more physical data collection.
Project website: https://yilongsong.github.io/CrowdMAPS/